

Developing Dashboards

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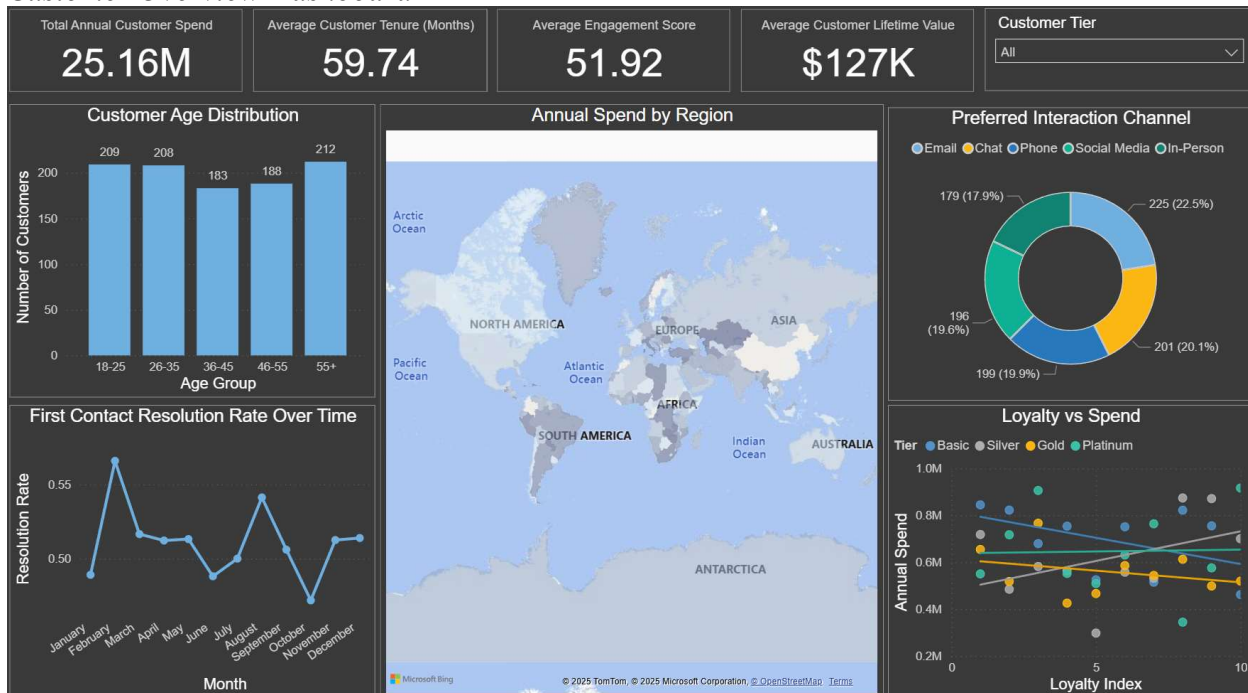
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Developing Dashboards

This Power BI dashboard was developed to demonstrate how interactive business intelligence tools can support deeper customer analysis and exploratory decision-making. Using the Business Innovation Intelligence dataset from Kaggle (Zahran, 2025), the dashboard integrates multiple customer dimensions into a single analytical view. As shown in Figure 1, high-level KPIs provide immediate context regarding customer value, engagement, tenure, and lifetime value, establishing a foundation for further investigation through supporting visualizations. The report was published to Power BI Service and the entire report page was pinned to an online dashboard, allowing the dashboard to function as an executive entry point while preserving full interactive capabilities. This approach ensures that users can seamlessly transition from the dashboard to the underlying report and continue to utilize dynamic filtering, cross-filtering, and other interactive features without loss of functionality.

Figure 1
Customer Overview Dashboard

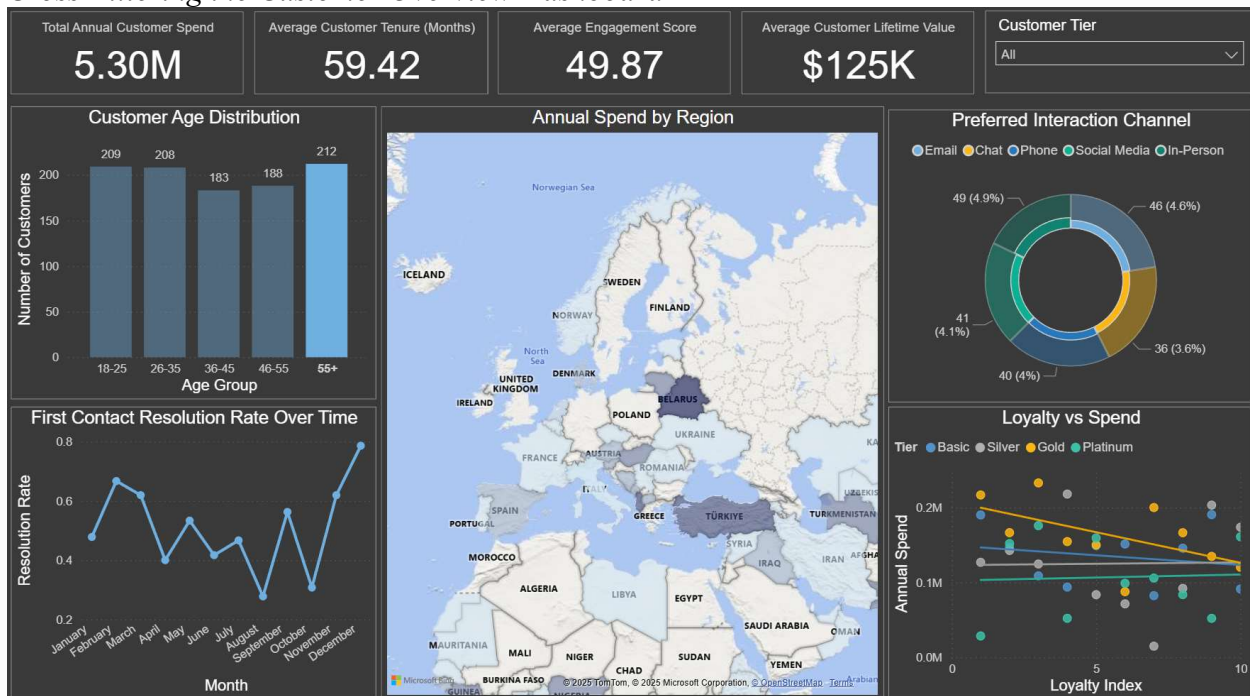


Note. This dashboard presents a customer-focused business intelligence overview, including total annual customer spend, average customer tenure, engagement score, and customer lifetime

value. Visualizations display customer age distribution, geographic spending patterns, preferred interaction channels, loyalty versus spend relationships, and first contact resolution rate over time.

Rather than relying heavily on static filters, the dashboard emphasizes interactivity through cross-filtering and dynamic filtering. Cross-filtering, illustrated in Figure 2, enables users to explore how customer behaviors and outcomes shift when a specific demographic segment is selected. For example, selecting the 55+ age group reveals differences in spending patterns, interaction preferences, and loyalty trends that may not be apparent at an aggregate level. Dynamic filtering, shown in Figure 3, allows users to narrow the analysis to specific customer tiers and interaction channels, supporting targeted evaluation of high-value segments. This layered filtering approach reflects real-world BI usage, where analysts iteratively refine views to uncover meaningful patterns.

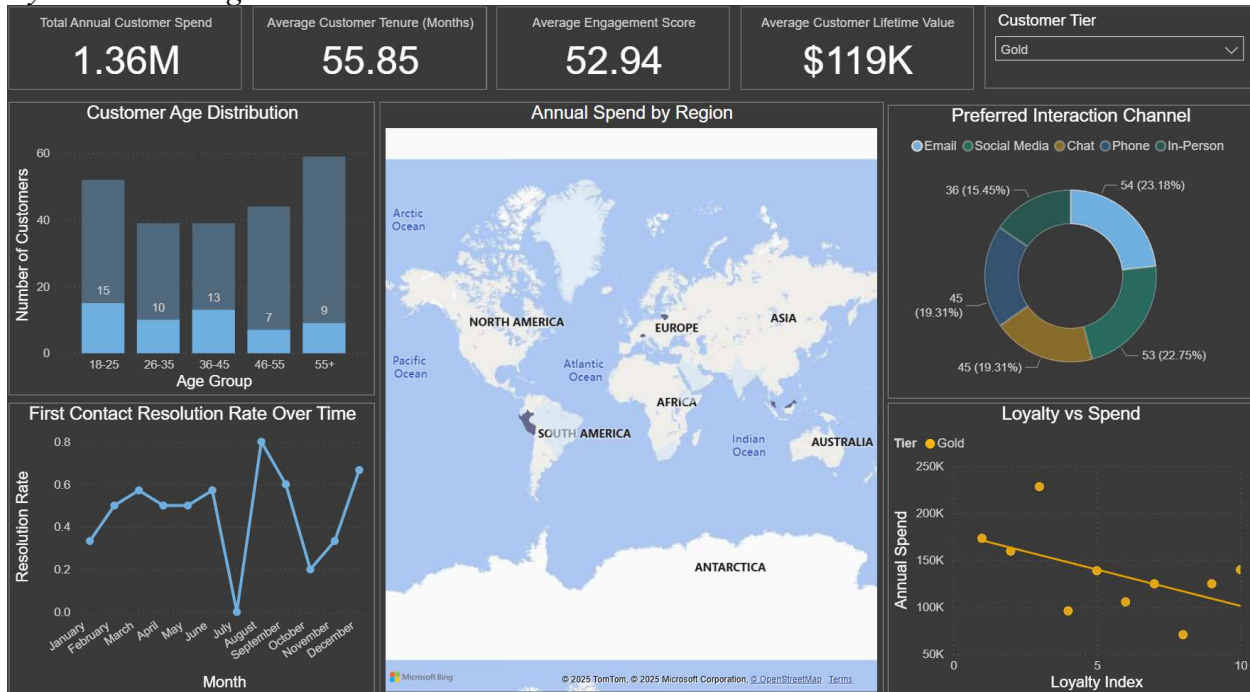
Figure 2
Cross Filtering the Customer Overview Dashboard



Note. This figure illustrates cross-filtering within the customer overview dashboard. Selecting the 55+ age group in the Customer Age Distribution chart dynamically updates all other visuals, including key performance indicators, geographic spending patterns, preferred interaction channels, loyalty versus spend relationships, and first contact resolution trends. This interaction

demonstrates how selecting a data element in one visualization filters related data across the dashboard in real time, enabling users to explore how customer behavior and value differ by demographic segment.

Figure 3
Dynamic Filtering the Customer Overview Dashboard

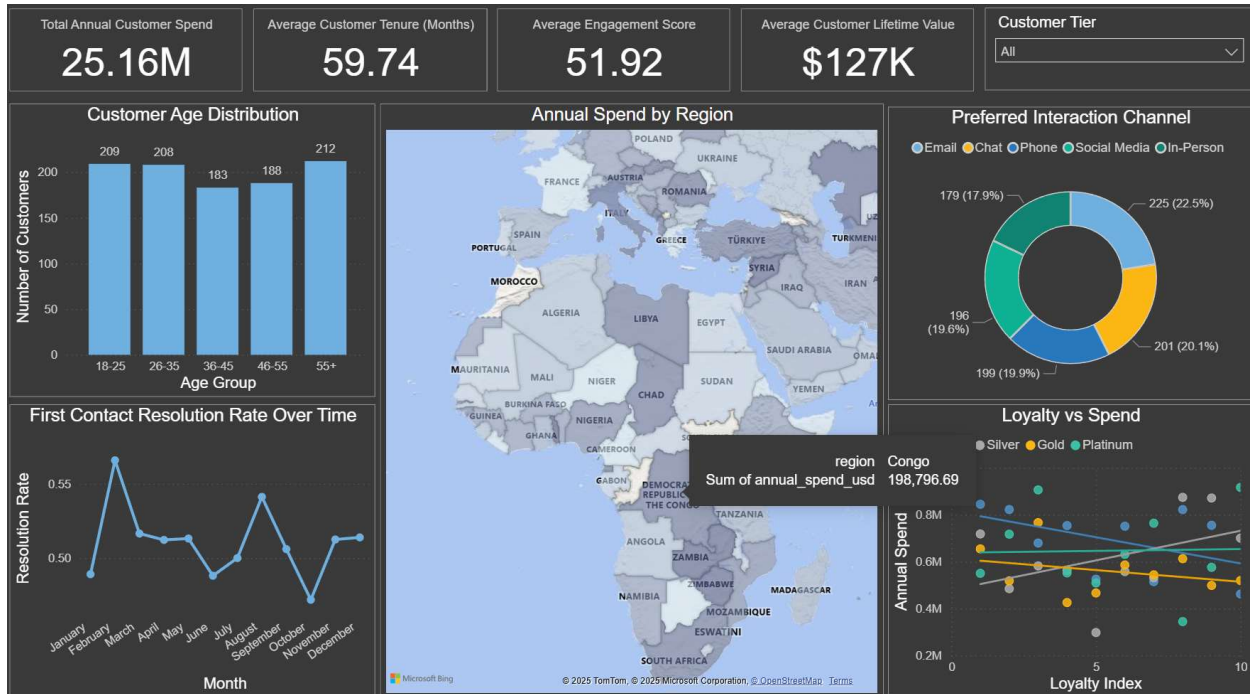


Note. This figure demonstrates dynamic filtering within the customer overview dashboard. The Customer Tier slicer was used to filter the dataset to Gold-tier customers, resulting in immediate updates to all key performance indicators and visualizations. Additional filtering was then applied by selecting the Email segment within the Preferred Interaction Channel chart. This further refined the dashboard to display customer behavior and performance metrics specific to Gold-tier customers who prefer email interactions. These combined interactions dynamically update all visuals in real time, illustrating how multiple filters can be applied simultaneously to support targeted customer analysis and exploratory decision-making.

User-friendly interactivity is further enhanced through tooltips, as demonstrated in Figure 4, which provide contextual metrics without disrupting the overall dashboard layout.

Collectively, these design choices illustrate how interactive dashboards transform raw customer data into actionable insights that inform segmentation strategies, retention efforts, and customer experience optimization.

Figure 4
Interactive Tooltip Displaying



Note. This figure demonstrates the use of interactive tooltips within the geographic visualization. Hovering over the Congo region displays a tooltip showing the total annual customer spend for that region without applying a filter or altering other dashboard visuals. Tooltips provide on-demand contextual information while preserving the overall dashboard view, enhancing usability and enabling users to quickly access detailed metrics without disrupting their analytical workflow.

Conclusion

In conclusion, this Power BI dashboard demonstrates how thoughtfully designed interactive visualizations can enhance business intelligence and support informed decision-making. By combining dynamic filtering, cross-filtering, and user-friendly interactivity, the dashboard enables users to explore customer data from multiple perspectives and uncover meaningful patterns related to customer value, behavior, and service performance. The integration of interactive visuals and an online dashboard structure illustrates how BI tools can move beyond static reporting to support exploratory analysis and strategic insight. Overall, this

project highlights the importance of intentional dashboard design in transforming complex datasets into actionable information that supports customer-focused business strategies.

References

Zahran, S. (2025). *Business innovation intelligence dataset* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/sarazahran1/business-innovation-intelligence>